

Case Study Assignment: Machine Learning 2

Fraudulent Claim Detection

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- **Introduction**

This study summarizes the analysis and predictive modelling performed to detect potentially fraudulent insurance claims. The goal was to create machine learning models that used historical claim data and consumer characteristics to categorize claims as real or fraudulent. Two models, Logistic Regression and Random Forest, were created and assessed using performance measures.

- **Problem Statement**

Global Insure faces significant monetary losses as a result of delayed detection of fraudulent insurance claims. The present fraud detection procedure is mainly based on manual inspections, which are inefficient and reactive. The company wants a proactive, data-driven solution to identify fraudulent or valid claims early in the approval process. The goal is to reduce financial risk, speed claims processing, and improve operational efficiency.

- **Business Objectives**

The key business objectives were –

- Build a prediction model based on historical claims data and consumer profiles and using the features in the data set to identify fraud patterns
- Enable early fraud detection by scoring new claims with a probability of fraud likelihood before approval
- Generate actionable insights that highlight the most predictive fraud indicators, supporting strategic decisions and continuous process improvement

- **Data Preparation**

The following steps were undertaken to prepare the dataset for modelling:

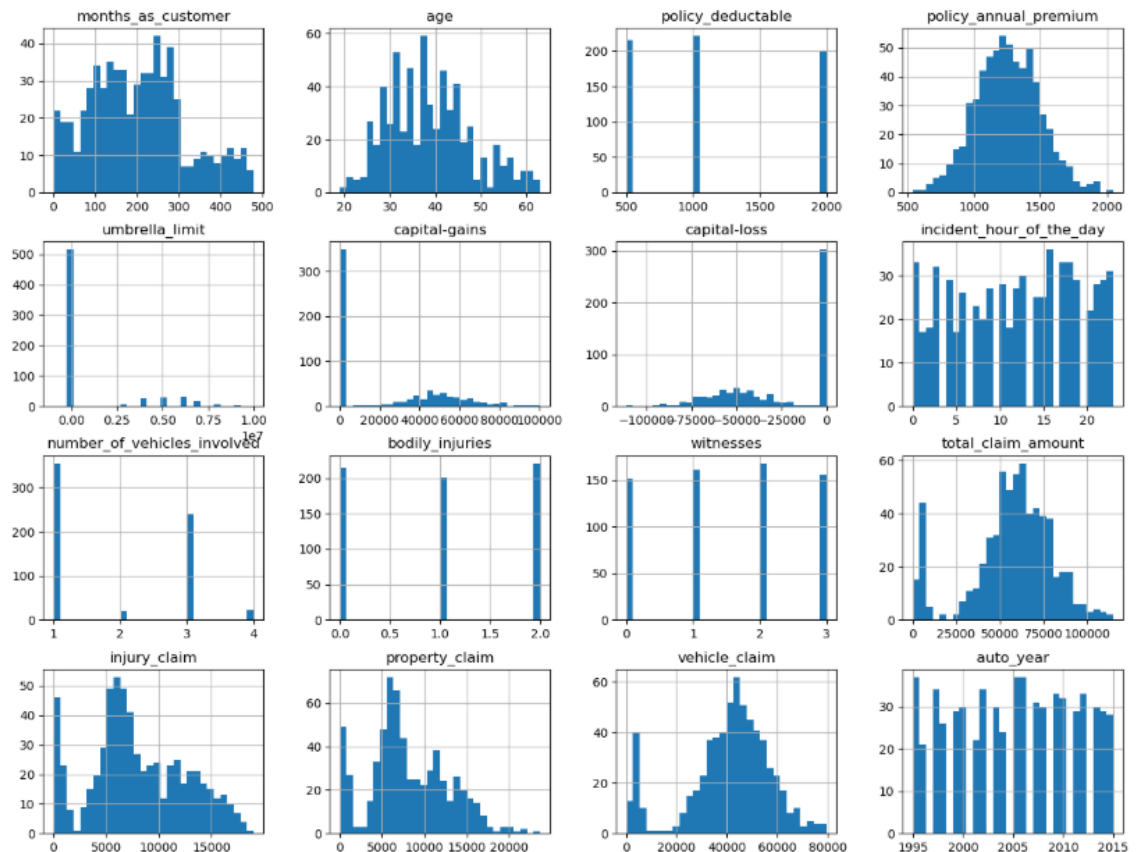
- Loaded the dataset and conducted an initial examination of data types, column distributions, and unique value counts.
- Identified and imputed or removed records with missing values in columns like 'authorities_contacted'.
- Dropped the column '_c39' which contained 100% missing values.

- Dropped high-cardinality identifier columns (e.g., policy_number, incident_location) to avoid overfitting.
- Removed rows with negative values in fields like 'capital-loss' where only non-negative values are logical.
- Converted date fields (e.g., 'incident_date') to datetime.
- Created dummy variables for categorical columns using get_dummies().
- Rescaled numeric features and balanced the dataset using RandomOverSampler to handle class imbalance.

• Exploratory Data Analysis

Key Insights from the Exploratory Data Analysis

• Univariate Analysis



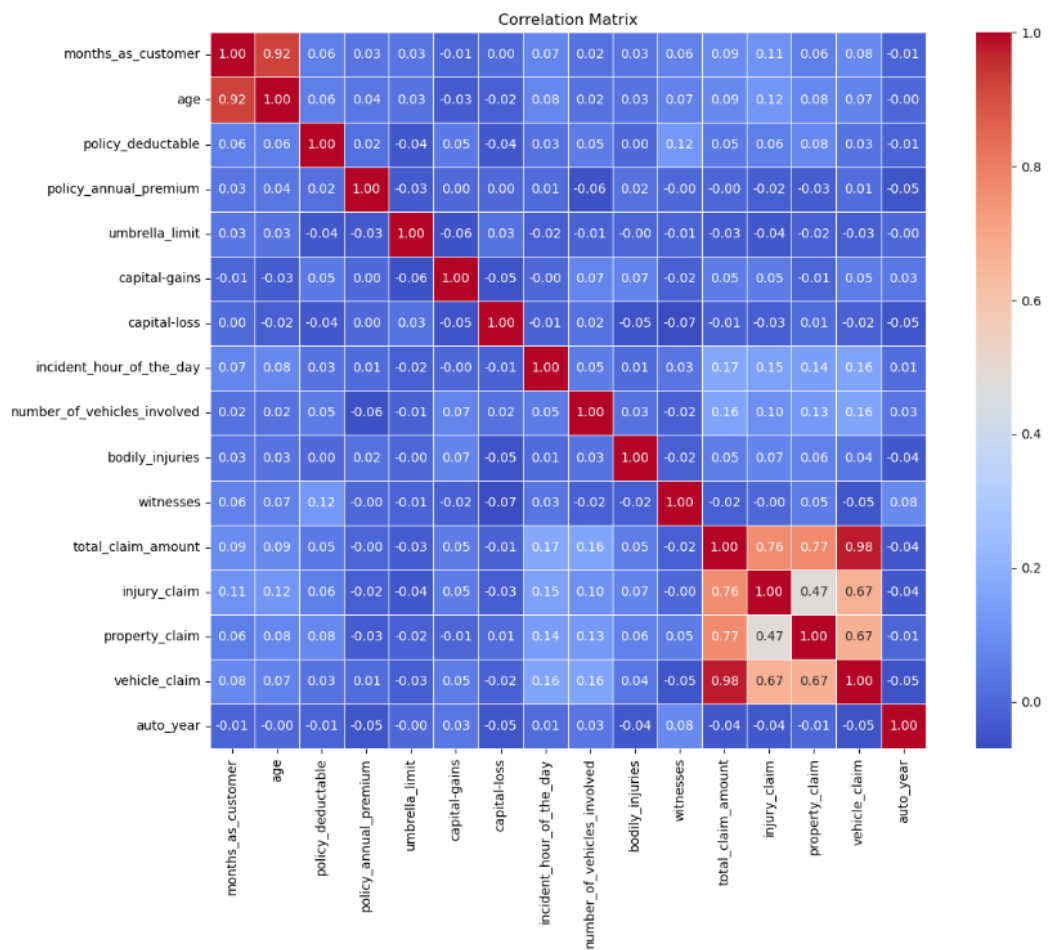
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○ Inferences from the univariate analysis are as under -

- The features that have an imbalanced or skewed distribution
 - umbrella_limit - most customers have a limit of '0' (right skewed)
 - capital_gains - values clustered around '0' (right skewed)
 - capital_loss - left skewed (values around '0')

- others where we can see some mild right skew
- injury_claim
- property_claim
- vehicle claim
- The following features have discrete peaks exhibiting category like behaviour
 - number_of_vehicles_involved
 - bodily_injuries
 - witnesses
 - policy_deductable - only 3 values
 - auto_year: Limited set of years
- The following features are normally distributed
 - policy_annual_premium
 - total_claim_amount

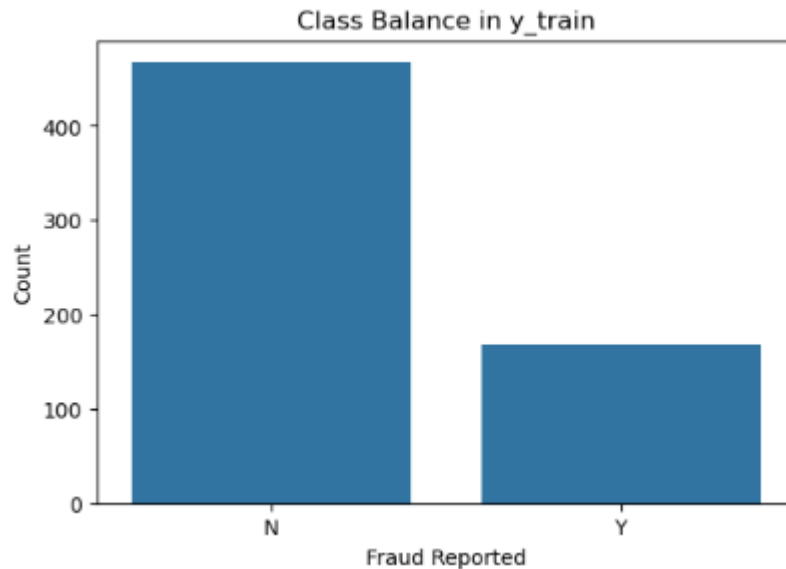
• Correlation Analysis



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- The following features are having a strong positive correlation
 - total_claim_amount & vehicle_claim - 0.98

- months_as_customer & age - 0.92
- total_claim_amount & injury_claim - 0.76
- total_claim_amount & property_claim - 0.77
- property_claim & injury_claim - 0.67

- **Class Imbalance**



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- We do observe a class imbalance which is not quite unusual in a problem like this.

- **Bivariate Analysis**

- Key interpretations of the bivariate analysis were –
 - insured_sex - Males (32%) are more likely to report fraud than females (23%)
 - insured_education_level - JD(29%), Associate & MD holders show slightly higher fraud rates. High school is the lowest (20%)
 - insured_occupation - Fraud more likely in executive-managerial(40%) and farming-fishing(38%) roles. They are lowest in admin, other service (17%)
 - insured_hobbies - Hobbies like chess (84%), cross-fit (83%), yachting (39%) show unusually high fraud risk.
 - policy_state - Indiana (30%) and Ohio (28%) have higher fraud rates vs. Illinois (22%)
 - auto_make - Audi, Chevrolet, and Mercedes ($\geq 31\%$) lead in fraud; Jeep is lowest (14%)
 - incident_type - Single Vehicle Collision (29%) is more fraud-prone than Vehicle Theft (14%)
 - collision_type - Rear and Front Collisions ($\sim 30\%$) more fraudulent than Side or unknown

- incident_severity - Major Damage shows extremely high fraud likelihood (61%) vs. Total Loss and Minor Damage (~12%)
- authorities_contacted - Other and Ambulance lead to more fraud cases (32%, 29%) compared to Police (21%)
- incident_state - Ohio (53%) and Pennsylvania (38%) have highest fraud likelihoods

• Feature Engineering

To enhance the model's ability to capture meaningful patterns and improve predictive accuracy, the following feature engineering steps were carried out:

- Derived binary indicators from categorical fields to represent fraud-prone categories (e.g., hobbies like 'chess' or 'cross-fit', severity levels).
- Consolidated low-frequency categories to reduce noise and dimensionality.
- Dropped columns with minimal variance or low predictive contribution as identified via correlation analysis and domain logic.
- Applied RandomOverSampler to synthetically balance the dataset, ensuring the fraud class was sufficiently represented for training.
- Performed one-hot encoding using `pd.get_dummies()` to convert categorical features into numerical format.
- Applied feature scaling to normalize numerical variables and maintain consistency across algorithms sensitive to magnitude.

• Model Building & Evaluation

Two supervised machine learning models—Logistic Regression and Random Forest—were built and evaluated using stratified cross-validation and a separate validation dataset. Key steps and outcomes are summarized below:

- Logistics Regression
 - **Feature Selection:**

- Utilized Recursive Feature Elimination with Cross-Validation (RFECV) to identify the most relevant predictors from the encoded training data.
- Final selected features included categories such as 'incident_severity' and 'insured_hobbies' which were statistically significant indicators of fraud.
- **Model Fitting and Interpretation:**
 - A logistic regression model was trained on the selected features, with multicollinearity assessed using Variance Inflation Factor (VIF).
 - The model was evaluated using statsmodels, allowing examination of coefficients, p-values, and confidence intervals.
- **Threshold Tuning and Evaluation:**
 - An optimal cutoff of 0.4 was determined using ROC and precision-recall trade-off analysis.
 - Performance on validation set:
 - **Accuracy:** ~86%
 - **Precision:** ~83%
 - **Recall (Sensitivity):** ~89%
 - **F1 Score:** ~0.86
- **Random Forest**
 - **Feature Importance and Selection:**
 - A baseline Random Forest model was trained to extract feature importances, and top features were retained for final modelling.
 - Features such as 'incident_severity', 'authorities_contacted', and 'insured_occupation' ranked highly.
 - **Model Tuning and Training:**
 - Hyperparameters (e.g., n_estimators, max_depth, min_samples_split) were tuned using GridSearchCV to enhance generalization.
 - The model was retrained using optimal parameters.
 - **Final Evaluation:**
 - The tuned model was evaluated on the validation set:
 - **Accuracy:** ~91%
 - **Precision:** ~87%
 - **Recall:** ~89%

- **F1 Score:** ~0.89

Overall, the Random Forest model outperformed Logistic Regression in accuracy and balance of precision-recall, making it the recommended choice for production deployment.

- **Insights & Recommendations**

- Fraudulent claims exhibit identifiable patterns linked to incident severity, customer hobbies, and occupation types, making predictive modelling highly viable.
- The Random Forest model is recommended for operational deployment due to its superior accuracy (~91%), balanced precision and recall, and robustness after hyperparameter tuning.
- Although the Logistic Regression model showed slightly lower performance (~86% accuracy), it remains valuable for its interpretability, especially in regulatory and audit contexts.